

The coupling process of awareness and disease under the influence of synergistic-hedging effect of information resources

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ABSTRACT

Information serves as a crucial yet invisible resources and can be categorized into two types: positive information resources and negative information resources. Different types of information resources interact to produce synergistic and hedging effects, which in turn influence individuals' adoption of targeted protective behaviors. These effects shape the scope and effectiveness of disease control measures and ultimately the dynamics of disease transmission. Additionally, the availability of medical resources enhances both individuals' overall resilience and the efficiency of disease response. To address these complexities, we propose a coupled model of awareness diffusion and disease transmission that incorporates the synergistic-hedging effect of information resources and influence of medical resources on disease containment. Using the microscopic Markov chain approach (MMCA), we derive the disease transmission threshold. Monte Carlo simulations reveal that the hedging effect of negative information resources on positive information resources contributes to an increased scale of infection, especially in the early stages of an outbreak. Conversely, the synergistic effects of positive information resources become more effective in inhibiting disease transmission as the sensitivity of information resources increases. Although this sensitivity does not affect the outbreak threshold, it significantly reduces the scale of infection. Notably, while increasing sensitivity to information resources gradually reduces the infection scale, enhanced the utilization of medical resources exerts a significantly stronger effect, rapidly and substantially lowering infection scale. Moreover, a combined strategy involving both information and medical resources proves to be the most effective in controlling disease outbreaks.

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1. Introduction

Infectious diseases pose a significant threat to individual health and well-being (Husain et al., 2025, pp. 28–50; Mao et al., 2025; Tian et al., 2025). Furthermore, they exert a profound impact on society, influencing economic stability and overall social welfare (Panda et al., 2024; Park & Koch, 2024; Samsudin et al., 2024). Therefore, it is crucial to comprehend the mechanisms of disease transmission and analyze the factors influencing it to effectively control infections.

Many classical disease transmission models have been proposed to describe the process of disease transmission in the population, providing a solid theoretical basis for studying the mechanism of disease transmission (Bailey, 1975; Hethcote, 2000; Kermack & McKendrick, 1927; Roy et al., 1982). Based on these classical disease models, some scholars have further considered a variety of factors affecting transmission, such as emotional heterogeneity, individual activity, individual mobility, and vaccination. You et al. introduced three emotional heterogeneity indicators and found that when the heterogeneity of individual emotional threshold was high and the heterogeneity of emotional response to information was low, disease transmission would be more easily controlled (You et al., 2023). Zeng et al. studied the impact of the correlation between activity and attractiveness on both epidemic spreading and rumor diffusion, finding that nodes with higher products of activity and attractiveness in time-varying networks significantly enhance disease spread while markedly inhibiting rumor diffusion (Zeng et al., 2023). Zhou et al. examined how individual mobility influences epidemic transmission, finding that individual mobility's influence on epidemic transmission depends on the distribution of the epidemic network (Zhou, Ma, et al., 2024). Li et al. summarized a multi-network model analysing how epidemic is influenced by multi-dose vaccine administration, revealing that receiving multiple doses significantly reduces epidemic transmission compared to single-dose acceptance (Li, Dong, et al., 2024). These extensions and improvements enable us to better understand the complexity of disease transmission and thus provide an important scientific basis for formulating effective prevention and control strategies.

Information is also an important factor affecting the spread of disease (Bao & He, 2025; Han et al., 2025; Huang et al., 2024; Wang et al., 2024, 2025; Yu & Huo, 2025). Some scholars even have further dissected the impact of information duality on disease transmission (Huo et al., 2022; Liu et al., 2023; Zhang & Wang, 2024). Du et al. discovered that during emergencies, competitive information transmission and interaction dynamics in double-layer networks expedite and broaden information dissemination compared to single-layer networks (Du & Fan, 2024). Fang et al. showed that there is an interaction between conflict views and epidemic transmission in multiple networks and individuals should strive to improve information literacy and actively seek out positive information (Fang et al., 2023). Li et al. established a $UA_1A_2U-SEIR$ (unaware, aware of positive prevention information and aware of negative prevention information-susceptible state, exposure state, infection state, and recovery state) model and found that promoting the rapid spread of positive information while curbing the spread of negative narratives can significantly reduce the spread of epidemics (Li et al., 2023, 2024b). Hu et al. aimed to investigate how active and passive information diffusion influences epidemic transmission within a higher-order network, highlighting that disseminating positive and negative prevention messages can significantly affect critical thresholds (Hu et al., 2024). Zhang et al. explored the interplay of multiple types of information on multiplex networks, taking into account both competitive and cooperative dynamics, suggesting that the pairwise interaction between each pair of information types strongly influences their final outbreak scale and diffusion threshold (Zhang et al., 2022).

Besides information dissemination, rational distribution of resources also has also been shown to be effective in disease control (Chen et al., 2018, 2019; Li et al., 2019; Lokhov & Saad, 2017; Yuan et al., 2023). Insufficient emergency medical resources can even worsen the spread of disease, so we can't ignore the importance of resources in disease control (Amado et al., 2023; Chen et al., 2025). Zhou et al. used a new three-layer co-evolutionary "information-epidemic-resources" model to study the interaction effects of information diffusion and virus transmission on the demand for medical resources. The results reveal that strengthening self-protection can reduce the fluctuation of individual demand, and effective restrictive protection can reduce the fluctuation of centralized resource storage (Zhou, Zhang, & Yang, 2024). Wang et al. integrated individual medical resource factors and public medical resource factors into SIR epidemic model and showed that greater magnitudes of both types weaken disease transmission (Wang et al., 2023). Zhao et al. proposed a strategy of distributing materials from surrounding areas to the epicenter in view of the serious shortage of supplies in the epicenter, and established a supply-side selection model considering the impact of the epidemic and the supply situation of candidate supply-side areas (Zhao et al., 2023). Zhu et al. investigated how asymptomatic infections affect disease transmission under information-driven resource allocation, showing that enhancing resource efficiency can shift the shorter incubation period from a first-order phase transition to a secondary phase (Zhu et al., 2022).

Previous research indicates that positive information diffusion effectively mitigates disease, whereas negative information dissemination can conversely facilitate disease spread. However, most existing studies have primarily focused on the effects of either positive or negative information on disease transmission in isolation, neglecting the potential interactions between them. Importantly, information acts as a vital yet often invisible resource which can be categorized into two types: positive information resources like medical service information and prevention and control measures; and negative information resources like misinformation about disease treatments and exaggerated reports that cause public panic. The greater the amount of positive information resources, the more likely individuals are to adopt appropriate protective behaviors; conversely, the greater the amount of negative information resources, the more likely individuals are to engage in inappropriate or absent protective responses. What is more, the two types of information resources can interact and

generate synergistic and hedging effects (Sun et al., 2024), that influence individuals' adoption of targeted protective behaviors. Similar information resources can reinforce and complement each other, creating a sense of consistency that enhances individuals' trust in these information resources and produces a synergistic effect in prompting them to take protective actions. Conversely, conflicting types of information resources may negate each other's influence, and the hedging effect among these conflicting resources can affect individuals' decisions to take protective actions. Overall, these synergistic and hedging effects collectively determine the extent and efficacy of individuals' implementation of specific protective behaviors, ultimately influencing the dynamics of disease transmission. However, limited research has addressed how these interactions affect disease spread. Besides, rational allocation of medical resources has proven to be crucial for effective disease control. Yet, few studies concentrated on disease transmission under the joint influence of information resources and medical resources, highlighting an important gap in the current understanding of disease control strategies. So, we focus on the influence of synergistic-hedging effect of information resources, propose a coupled UA_1A_2U –SIS model to study the impact of information resources and medical resources on diseases. By addressing this gap, we aim to offer valuable insights that could enhance the effectiveness of public health interventions.

The innovative points of this article are as follows:

- (i) It pioneers the conceptualization of information as a dual-type resource (positive/negative) whose synergistic-hedging interactions—not merely their individual effects—govern protective behavior adoption. Most existing literature examines the one-way influence of positive/negative information on disease transmission as independent variables. This study innovatively proposes that these two types of information resources do not simply add up, but instead interact dynamically through a synergy effect (where similar information reinforces each other) and a counteraction effect (where different types of information are mutually exclusive), jointly shaping individual protective behavior decisions.
- (ii) It bridges a critical gap by integrating information and medical resources into a unified coupled awareness-disease framework, revealing their bidirectional feedback mechanisms. Previous studies either focused on the dynamics of information dissemination or on the strategies for allocating medical resources, but rarely explored the joint influence mechanism of these two types of resources.

The remainder of this paper is arranged as follows. We describe the multiple networks model in Section 2. In Section 3, we conduct a theoretical analysis of the spreading dynamics using the MMCA method. To verify the theoretical predictions made in Section 3, we run numerical simulations in Section 4. Finally, we conclude our work in Section 5.

2. The awareness-disease coupled model considering resources

Information serves as crucial yet invisible resources, which can also be categorized into two types: positive information resources and negative information resources. Positive information resources encompass medical service details, disease updates, prevention and control measures, health education materials, and official disease prevention and control guidelines. Conversely, negative information resources include statements like “COVID-19 is trivial, no different from a common cold” and “Vaccination is unnecessary, it's ineffective and may even cause side effects”. Different types of information resources can interact and generate synergistic or hedging effects. To be specific, on the one hand, information resources of the same type can confirm each other and thus have a synergistic effect on individual behavior. Conversely, conflicting information types may negate each other's influence. For example, when individuals are exposed to positive information resources about the COVID-19 vaccine in their surrounding environment, such as neighbors generally sharing that vaccination can effectively prevent COVID-19 infection and reduce transmission rates, these positive information resources mutually reinforce and validate each other, forming a synergy effect that enhances individuals' recognition and understanding of the positive value of the vaccine. Thereby, it further promotes their adoption of vaccination to protect their own health. On the contrary, when an individual simultaneously receives two types of contradictory information resources: on the one hand, someone shares positive information such as the effectiveness and safety of the vaccine; on the other hand, someone spreads negative information about the side effects or potential risks of the vaccine. There will be a hedging effect between these two types of information resources, leading to conflicts and uncertainties in the individual's cognition. Thereby weakening their willingness to get vaccinated. These synergistic or hedging effects define the extent and effectiveness with which individuals adopt targeted protective behaviors, thereby directly influencing disease transmission dynamics.

Additionally, medical resources serve not only to treat diseases such as medicines and vaccines but also to prevent them during early outbreaks including masks, protective clothing, and goggles, which also have an impact on disease. For example, Fig. 1 shows the vaccination status of the population and the number of people infected with diseases at the same time. The data reveal a clear inverse relationship between vaccination coverage and infection incidence: in early 2021 (January–February), low daily vaccination uptake (<0.05% of the population) coincided with the first major epidemic wave (>700 cases per million per day). As coverage increased through mid-2021, infection incidence declined progressively, reaching its lowest level (<150 cases per million per day) during November–December 2021 following the peak vaccination rate (~0.25% per day) observed in September 2021. This temporal pattern demonstrates that increased vaccine coverage is associated with reduced disease transmission, while limited vaccination access correlates with larger epidemic outbreaks.

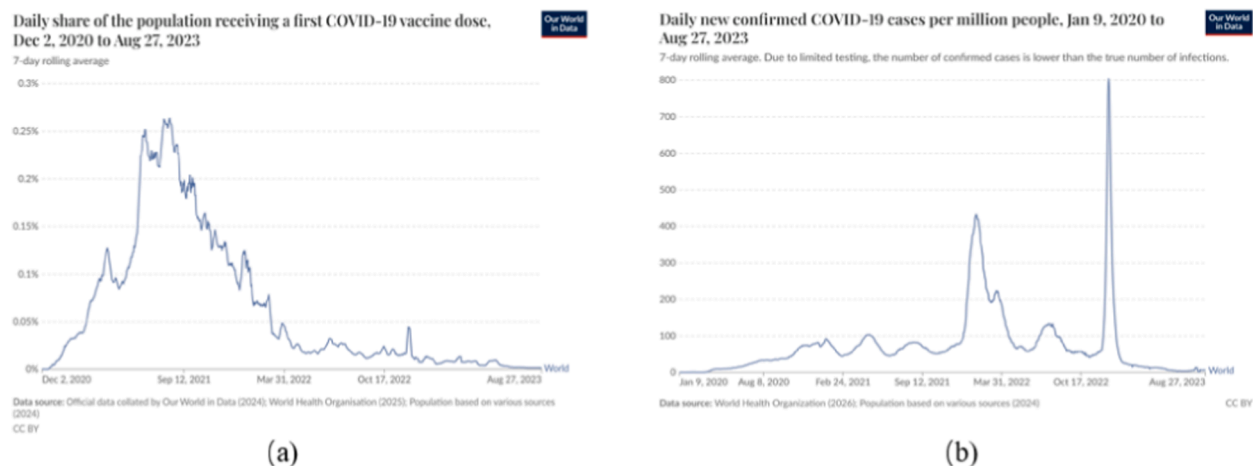


Fig. 1. (a) Daily share of the population receiving a first COVID-19 vaccine dose, Dec 2, 2020 to Aug 27, 2023. (b) Daily new confirmed COVID-19 cases per million people, Jan 9, 2020 to Aug 27, 2023. Data source: <https://ourworldindata.org/>.

To offer a more precise depiction of this phenomenon, we construct a network to portray awareness diffusion and disease transmission process under the influence of synergistic-hedging effect of information resources and medical resources as shown in Fig. 2. There are three kinds of states of awareness diffusion layer in A: Unaware the disease (U), Correct perception of disease (A_1) and aware the disease but believe the disease poses no threat and is not dangerous, misperception about disease (A_2). Two kinds of states of disease transmission layer in B: Individuals who are healthy (S) and who have been

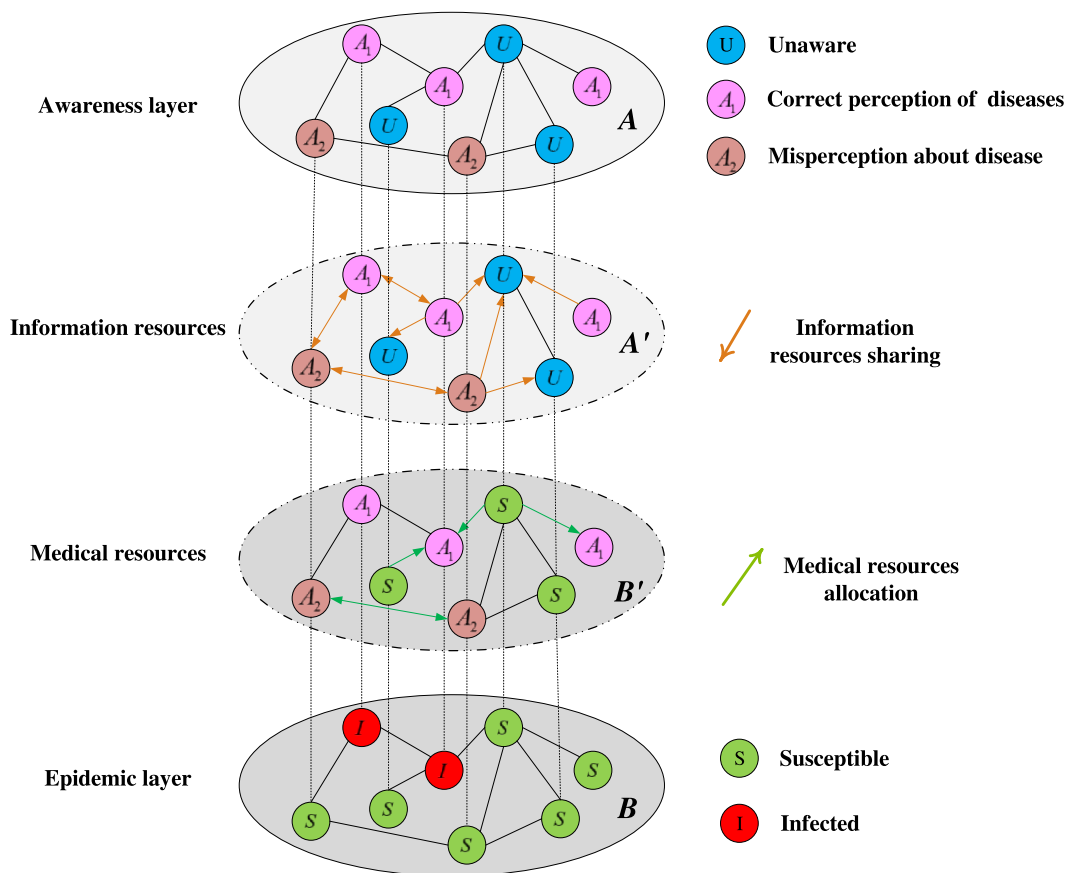


Fig. 2. Structure of awareness-disease coupled model considering the influence of information resources and medical resources. There is a one-to-one correspondence between nodes at each layer.

infected by the disease (I). The solid lines represent the awareness and disease layers, while the dotted line in the middle indicates information resources sharing and medical resources allocation. Specifically, as people increasingly communicate online and donate medical resources that require in-person contact, the network topology remains consistent for both the awareness and information layers, as well as the disease and medical resource layers. The orange and green arrows symbolize the processes of information resource sharing and medical resource allocation, respectively.

The network model propagation process is assumed as follows:

Assumption 1. Awareness diffusion

We use a UA_1A_2U model to delineate the transitions between individual awareness as shown in Fig. 3. These states capture fundamentally different awareness states that drive divergent behavioral responses during epidemics. U means that the individuals have no knowledge of the disease's existence or outbreak. For example, during the early phase of the 2020 COVID-19 pandemic, residents in isolated communities who received no media coverage remained unaware of the emerging threat. Consequently, they took no protective measures simply due to information absence—but once accurately informed, they readily adopted masks and social distancing. A_1 means that individuals are aware of the danger of the disease. Individuals possess accurate risk perception and recognize the disease's danger. These individuals possess accurate risk perception, follow official health advisories, understand transmission risks, and consistently practice evidence-based protections (e.g., mask-wearing, hand hygiene), thereby reducing their infection probability. A_2 means that the individuals are aware of the disease but hold false beliefs that distort risk assessment. A representative case: during the same pandemic, some individuals acknowledged COVID-19's existence yet believed “the virus only affects the elderly” or “drinking alcohol kills the virus”. This led them to actively reject effective protections or adopt harmful pseudo-protective behaviors. Epidemiologically, U individuals pose a unidirectional risk: they are vulnerable to infection due to passive exposure, yet they do not actively amplify transmission through harmful behaviors. Conversely, A_2 individuals generate a bidirectional risk: (i) they face elevated personal infection risk due to rejected protections, and (ii) they actively amplify community transmission through erroneous behaviors (e.g., alcohol-consuming gatherings justified as “virus-killing rituals”).

So we suppose individuals in U state can turn to A_1 or A_2 state with a probability λ_1 or λ_2 . And individuals in A_1 or A_2 state can also convert to U state with a probability δ_1 or δ_2 after a certain time period. It is worth noting that due to herd mentality and neighboring state influence, individuals may oscillate between A_1 and A_2 state. To model this phenomenon, we assume that the individual state transition follows a threshold triggering mechanism (T_i). Specifically, for an A_1 individual, when the proportion of neighbors in A_2 state and U state around individual A_1 exceeds the conversion ratio φ_1 ($0 \leq \varphi_1 \leq 1$), the transformation process can be initiated ($T_i^{A_1} = 1$). On this basis, the probability η_1 of individual A_1 transform to A_2 has an exponential function relationship with the number of A_2 and U state neighboring individuals. If the ratio doesn't exceed φ_1 , the threshold triggering mechanism is not activated ($T_i^{A_1} = 0$), and the individual's state remains unchanged. As well, when the proportion of neighbors in A_1 state around individual A_2 exceeds the conversion ratio φ_2 ($0 \leq \varphi_2 \leq 1$), individual A_2 will satisfy the transition threshold $T_{i(t)}^{A_2} = 1$ and turn into A_1 state with a probability of η_2 in the next time step. So we use $T_i^{A_1}$ and $T_i^{A_2}$ to express the threshold of individual transformation between A_1 and A_2 :

$$T_{i(t)}^{A_1} = H \left[\frac{\sum_j a_{ji} (P_{j(t)}^{A_2} + P_{j(t)}^U)}{\sum_j a_{ji}} - \varphi_1 \right] \quad (1)$$

$$T_{i(t)}^{A_2} = H \left[\frac{\sum_j a_{ji} P_{j(t)}^{A_1}}{\sum_j a_{ji}} - \varphi_2 \right] \quad (2)$$

$H(x)$ is a Heaviside function. When $x > 0$, both $T_i^{A_1}$ ($T_i^{A_2}$) equal 1, signifying the threshold triggering mechanism activates, causing the individual's state change. When $x < 0$, $T_i^{A_1}$ and $T_i^{A_2}$ equals 0, suggesting that the A_1 or A_2 state remain unchanged. Furthermore, the specific probability η_1 of converting A_1 to A_2 increases with the number of A_2 and U state neighboring individuals and the η_2 is also the same. As whole, η_1 and η_2 can be expressed as:

$$\eta_1 = \frac{1}{2} \eta_1^0 (1 + \pi) \frac{\sum_{ij} a_{ij} P_j^{A_2}}{\sum_{ij} a_{ij} T_{i(t)}^{A_1}} \quad (3)$$

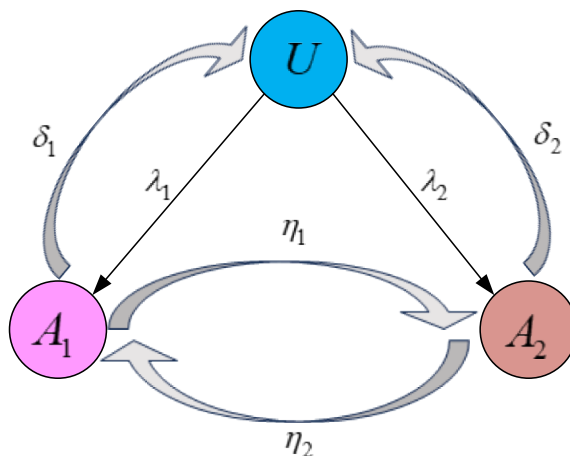


Fig. 3. Awareness diffusion process.

$$\eta_2 = \frac{1}{2} \eta_2^0 (1 + \pi) \frac{\sum a_{ij} p_i^{A_1}}{\sum a_{ij} T_{i(t)}^{A_2}} \quad (4)$$

where η_1^0 and η_2^0 are the initial transition probability, π indicates the intensity of neighboring influence. A larger π means that individual's transition between A_1 and A_2 is more likely to be influenced by their neighbors once the proportion of neighbors exceeds the conversion ratio. Fig. 4 Show the conversion probability η_1 and η_2 between A_1 and A_2 state when $T_i^{A_1} = T_i^{A_2} = 1$, $\eta_1^0 = \eta_2^0 = 0.35$.

When π is fixed, the probability η_1 of individual A_1 converting to A_2 increases with the number of neighboring neighbors A_2 , and when the number of neighboring neighbors A_2 is fixed, the probability η_1 of individual A_1 converting to A_2 increases with the neighbor influence intensity π . This trend is observed for η_2 .

Assumption 2. Disease transmission

We employ an SIS (Susceptible-Infected-Susceptible) model to characterize disease transmission dynamics, as illustrated in Fig. 5. This framework is specifically appropriate for pathogens that confer no lasting immunity upon recovery—such as common cold coronaviruses, gonorrhea, and *Chlamydia trachomatis*—where individuals immediately return to the susceptible pool after recovery (Chang et al., 2025; Gracy et al., 2025; Maojie, 2026; Zhu & Wang, 2026). When a healthy individual encounters an already infected individual, there is a probability β of infection. Infected individuals will recover with probability μ through treated or naturally rely on their own immune system. Critically, recovered individuals do not acquire permanent immunity and immediately re-enter the susceptible compartment, enabling recurrent infection cycles.

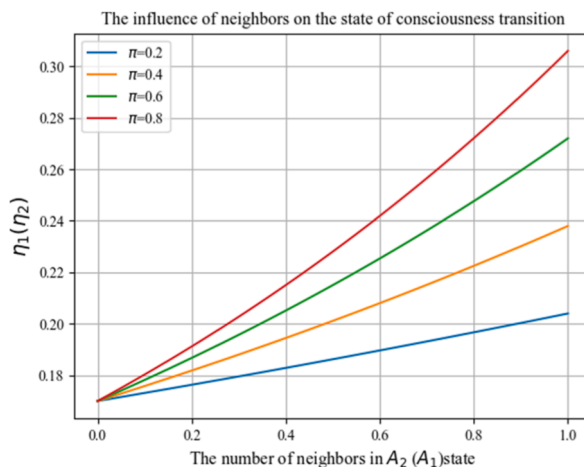


Fig. 4. The conversion probability between A_1 and A_2 state when $T_i^{A_1} = T_i^{A_2} = 1$, $\eta_1^0 = \eta_2^0 = 0.35$

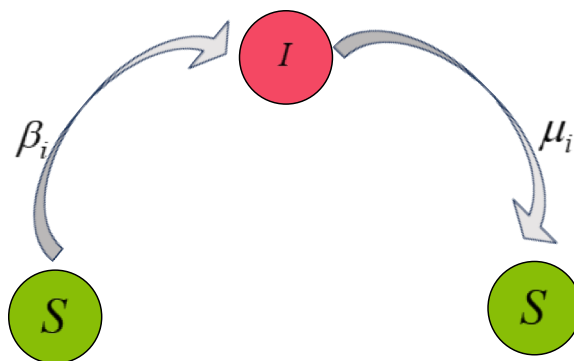


Fig. 5. Disease transmission.

Assumption 3. state coupling

In general, there are a total of 6 possible states: US , UI , A_1S , A_2S , A_1I and A_2I . We assume that once an individual is infected, the symptoms of the disease directly affect the daily life and work of the individual and make the individual aware of the danger of the disease. Thus, A_2I and UI are excluded from the model. It is worth noting that our assumption represents a deliberate simplification most applicable to acute infections with pronounced, disruptive symptoms, where the somatic feedback between infection and awareness is strong. In reality, however, this assumption carries inherent limitations, as a non-negligible fraction of infected individuals may maintain incorrect risk perceptions despite direct infection experience, particularly in cases of asymptomatic, mild infections or delayed symptom onset. Finally, there are four states: US , A_1S , A_2S and A_1I in our model. US refers to individuals who do not perceive the danger of disease and are susceptible, A_1S refers to individuals who perceive the danger of disease and are susceptible, A_2S refers to individuals who believe the disease poses no threat and is not dangerous and are susceptible and A_1I refers to individuals who perceive the danger of disease and are already infected.

Assumption 4. Information resources sharing and reception

(1) Generation of information resources

Individuals' different perceptions of the disease will give rise to different types of information resources. We assume that A_1 state individuals will generate a certain amount of positive information resources based on their perception of the disease, such as medical service information, disease update and prevention and control measures, health education materials and official guidelines for disease prevention and control, etc. These will guide people to treat the disease correctly and take comprehensive appropriate measures to avoid infection. Although A_2 state individuals are aware of the disease, they believe that the disease is not dangerous and even does not cause a great threat. For example, during the Covid-19, some people think that "COVID-19 is no big deal, just like a common cold." "There is no need to vaccinate, vaccinations are useless and even have side effects." They also generate information resources at each time step based on their misperception of the disease. However, such information resources will induce individuals not to take protective measures or even take wrong protective measures, hindering disease prevention and control, thus they are defined as negative information resources in this paper. And we use the negative sign indicates the negative utility of such information resources. The U state individuals do not produce any information resources because they don't aware the disease. Due to individual heterogeneity and different health status, the amount of information resources generated by each individual is different. More connected individuals tend to produce greater volumes of information resources, so we measure the amount of information resources for each individual by a quantity related to the degree k_i . b ($b > 0$) represents the relationship between the amount of information resources and the degree of nodes. The larger b is, the greater the correlation between the amount of information resources owned by an individual and the degree. Besides, when the individual is infected, the urge to restore health stimulate individuals' self-learning willingness to explore more information resources. They will seek information resources about treatment through different channels such as Internet search or other means, which is equivalent to an additional way to acquire information resources. So infected individuals have more information resources than healthy individuals and they are related to the individual's own learning ability a . Thus, the amount of information resources self-generated by each individual is expressed by M_{0i} as follows:

$$M_{0i}^{A_1S}(t) = bk_i \quad (5)$$

$$M_{0i}^{A_2S}(t) = -bk_i \quad (6)$$

$$M_{0i}^{A_1I}(t) = b(k_i + a) \quad (7)$$

$$M_{0i}^{US}(t) = 0 \quad (8)$$

(2) Synergistic and hedging effects of information resources.

Individuals who own information resources (A_1 state and A_2 state) will share their information resources with neighbors in all states, and the quantity of information they possesses does not decrease during the sharing. Every individual (U, A_1 and A_2 state) all receive information resources shared by others. Different types of information resources can influence each other, producing synergistic and hedging effects that impact individuals' adoption of targeted protective measures. When individuals receive information resources are in line with those they generated based on their knowledge of a disease, mutual verification creates a synergistic effect that reinforces their beliefs and encourages them to take appropriate measures. Conversely, when they receive information resources that contradict, a counteracting hedging effect arises from mutual refutation, leading to indecision and diminishing their capacity to take protective measures. Overall, on the one hand, information resources of the same kind can confirm each other and thus have a synergistic effect on individual behavior, on the other hand, negative information resources have the potential to negate the positive effects of positive information resources, while conversely, positive information resources can also mitigate the adverse effects of negative information resources.

To sum up, we use Δ_i to represents the total amount of information resources that individual i received from neighbors, and $\Delta_i(t) = M_{i+} + M_{i-}$, which M_{i+} denotes the total amount of positive informational resources received from neighbors by individual i at time t , while M_{i-} represents the total amount of negative information resources received from neighbors. $M_i(t)$ denotes the total amount of information resources owned by individual i at time t including the self-generated information resources M_{0i} and information resources obtained from neighbors Δ_i . Therefore, the amount of information resources owned by each individual is as follows. To obtain closed-form expressions for the epidemic threshold and to clarify the mechanistic role of information resources without introducing additional nonlinear complexity. We set $b = 1$ in equations (5)–(7). It allows us to isolate the core dynamics of awareness-disease interaction before considering more complex functional forms.

$$\begin{aligned} M_i^{A_1S} &= M_{0i}^{A_1S} + \Delta_i^{A_1S} \\ &= k_i^{A_1S} + \Delta_i^{A_1S} \end{aligned} \quad (9)$$

$$\begin{aligned} \Delta_i^{A_1S}(t) &= M_{i+}^{A_1S} + M_{i-}^{A_1S} \\ &= \left(\sum a_{ij} P_j^{A_1S} k_j^{A_1S} + \sum a_{ij} P_j^{A_1I} (k_j^{A_1I} + a) \right) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \\ &= \sum a_{ij} (P_j^{A_1S} k_j^{A_1S} + P_j^{A_1I} (k_j^{A_1I} + a)) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \end{aligned} \quad (10)$$

$$\begin{aligned} M_i^{A_2S} &= M_{0i}^{A_2S} + \Delta_i^{A_2S} \\ &= -k_i^{A_2S} + \Delta_i^{A_2S} \end{aligned} \quad (11)$$

$$\begin{aligned} \Delta_i^{A_2S}(t) &= M_{i+}^{A_2S} + M_{i-}^{A_2S} \\ &= \left(\sum a_{ij} P_j^{A_1S} k_j^{A_1S} + \sum a_{ij} P_j^{A_1I} (k_j^{A_1I} + a) \right) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \\ &= \sum a_{ij} (P_j^{A_1S} k_j^{A_1S} + P_j^{A_1I} (k_j^{A_1I} + a)) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \end{aligned} \quad (12)$$

$$\begin{aligned} M_i^{A_1I} &= M_{0i}^{A_1I} + \Delta_i^{A_1I} \\ &= k_i^{A_1I} + a + \Delta_i^{A_1I} \end{aligned} \quad (13)$$

$$\begin{aligned} \Delta_i^{A_1I}(t) &= M_{i+}^{A_1I} + M_{i-}^{A_1I} \\ &= \left(\sum a_{ij} P_j^{A_1S} k_j^{A_1S} + \sum a_{ij} P_j^{A_1I} (k_j^{A_1I} + a) \right) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \\ &= \sum a_{ij} (P_j^{A_1S} k_j^{A_1S} + P_j^{A_1I} (k_j^{A_1I} + a)) + \sum a_{ij} P_j^{A_2S} (-k_j^{A_2S}) \end{aligned} \quad (14)$$

$$M_i^{US} = M_{0i}^{US} + \Delta_i^{US} = 0 + \Delta_i^{US} \quad (15)$$

$$\begin{aligned}
\Delta_i^{US}(t) &= M_{i+}^{US} + M_{i-}^{US} \\
&= \left(\sum a_{ij} P_j^{A_1 S} k_j^{A_1 S} + \sum a_{ij} P_j^{A_1 I} (k_j^{A_1 I} + a) \right) + \sum a_{ij} P_j^{A_2 S} (-k_j^{A_2 S}) \\
&= \sum a_{ij} (P_j^{A_1 S} k_j^{A_1 S} + P_j^{A_1 I} (k_j^{A_1 I} + a)) + \sum a_{ij} P_j^{A_2 S} (-k_j^{A_2 S})
\end{aligned} \tag{16}$$

Assumption 5. Medical resource allocation

(1) Generation of medical resources.

We assume that only individuals in healthy S state can produce r unit of medical resources, whereas those who are infected are unable to do so since they are unable to engage in work or production normally.

(2) Allocation and obtainment of medical resources.

Individuals in S state will allocate medical resources to their neighbors in A state who actively seek the protection of medical resources. To be specific, US state individuals will allocate all their resources because they haven't yet realized the disease. Based on cognitive homogeneity (He & Bauch, 2024; Kadelka & McCombs, 2021; Stefkovics et al., 2024), individuals tend to form connections, interact, and share medical resources with those who hold similar beliefs, risk perceptions, or worldviews. So A_2S state individuals only allocate medical resources to others who have the same opinion of disease (A_2). A_1S state individuals also donate medical resources to help like-minded neighbors (A_1) based on local infection densities and the proportion of A_1S individual donated medical resources α_i decreases with the local infection densities increases, denotes as:

$$\alpha_i(t) = (1 - \rho_i(t))\alpha_0 \tag{17}$$

where $\rho_i(t) = \frac{1}{k_i} \sum_{j \in \Omega_i} P_j^I$, α_0 represents the basic donation probability.

What's more, A_1 individuals will take the initiative to obtain medical resources from S individuals to seek protection from infection or rapid recovery after infection due to the perception of the risk of disease, while A_2 individuals and U individuals will not take the initiative to obtain medical resources. To sum up, the amount of medical resources owned by individuals in each state is as follows:

$$R_i^{A_1 S}(t) = 1 - \alpha_i(t) + \sum b_{ij} P_j^{A_1 S} \frac{\alpha_i(t)}{\sum_{l \neq i} b_{jl} P_l^{A_1} + 1} + \sum b_{ij} P_j^{US} \frac{1}{\sum_{l \neq i} b_{jl} P_l^{A_1} + 2} \tag{18}$$

$$R_i^{A_2 S}(t) = \frac{1}{\sum_j b_{ij} P_j^{A_2} + 1} + \sum b_{ij} P_j^{A_2 S} \frac{1}{\sum_{l \neq i} b_{jl} P_l^{A_2} + 2} \tag{19}$$

$$R_i^{A_1 I}(t) = \sum b_{ij} P_j^{A_1 S} \frac{\alpha_i(t)}{\sum_{l \neq i} b_{jl} P_l^{A_1} + 1} + \sum b_{ij} P_j^{US} \frac{1}{\sum_{l \neq i} b_{jl} P_l^{A_1} + 2} \tag{20}$$

$$R_i^{US}(t) = \frac{1}{\sum_j b_{ij} P_j^{A_1} + 1} \tag{21}$$

For individual i in the A_1S state, the equation's right side consists of three parts: the remaining medical resources after allocating the medical resources of part α_i , the medical resources obtained from the surrounding A_1S neighbors, and the medical resources obtained from the surrounding US neighbors. For individual i in the A_2S state, the right side of the equation includes the amount of medical resources left for themselves after allocating the medical resources to their A_2 neighbors and the medical resources obtained from the surrounding A_2S neighbors. For individual i in the A_1I state, they can only get medical resources from the neighbors, where the first term on the right represents the medical resources obtained from the A_1S individual, and the second term represents the medical resources obtained from the US individual. For individual i in the US state, the medical resource amount is the amount left after distributing medical resources to neighbors. Note that, we consider that the resources of S state node will not be accumulated at each time step. Namely, if an S state node has no I state neighbors at one time step, it will consume the resource in some way, and the amount of resources in the network will be zero at the beginning of the next time step.

Assumption 6. The impact of information resources on disease

The infection rate of an individual is related to the state of individual awareness and the amount of resources possessed by individuals which is consisted of information resources and medical resources. To be specific, we suppose that U state

individuals will not take any precautions to avoid infection because they are not aware the disease, while individuals in the A_1 state have lower infection risk by attenuation factor $\gamma_1 (0 \leq \gamma_1 \leq 1)$ and A_2 state have enhanced infection risk by enhancement factor $\gamma_2 (1 \leq \gamma_2 \leq 2)$ because they perceive the danger of disease or misperceive the disease. The more positive information resources an individual possesses, the more comprehensive protection measures will be taken, and the infection rate of an individual will be greatly suppressed. On the contrary, the more negative information resources an individual possesses, the greater the probability that an individual will not take protective measures or even take wrong protective measures will increase the possibility of infection. It is worth noting that when A_1 individuals receive more negative information resources, they may also take some wrong preventive measures or do not take preventive measures to increase the risk of infection. Similarly, when A_2 individuals receive a greater amount of positive information resources, they may also take appropriate precautions to reduce the risk of infection. Besides, the more protective medical resources an individual has, such as masks and disinfection supplies can be used for protection, the lower the probability of being infected. Therefore, we assume that the infection rate of individuals in various states is:

$$\beta_i^{A_1S}(t) = \frac{1}{2} \gamma_1 \beta \frac{1}{1 + e^{sM_i^{A_1S}(t)}} (1 - \varepsilon)^{R_i^{A_1S}(t)} \quad (22)$$

$$\beta_i^{A_2S}(t) = \frac{1}{2} \gamma_2 \beta \frac{1}{1 + e^{sM_i^{A_2S}(t)}} (1 - \varepsilon)^{R_i^{A_2S}(t)} \quad (23)$$

$$\beta_i^{US}(t) = \frac{1}{2} \beta \frac{1}{1 + e^{sM_i^{US}(t)}} (1 - \varepsilon)^{R_i^{US}(t)} \quad (24)$$

where β denotes the initial infection rate, s denotes individual's sensitivity to information resources, and $s \in (0, 1)$. We assume that $y = \frac{2}{1 + e^{sM_i(t)}}$ to present the influence of information resources on the infection rate of disease clearly as shown in Fig. 6. The sensitivity of individuals to information resources significantly influences the variation in their infection probability. When an individual exhibits low sensitivity (blue curve), the infection rate increases gradually and approaches its maximum when the total amount of negative information resources exceeds that of positive ones (moving left on the x-axis). Conversely, when positive information resources outweigh negative ones, the infection rate gradually decreases toward its minimum (moving right on the x-axis). In contrast, individuals with high sensitivity (red curve) respond more sharply to changes of information resources. Even a slight excess of negative information resources over positive information resources causes the infection rate to rapidly reach its maximum. Similarly, when positive information resources marginally surpass negative information resources, the infection rate quickly drops to its minimum. In summary, greater sensitivity results in a more abrupt transition of infection rate in response to shifts in the ratio of positive to negative information resources.

ε represent the utilization rate of medical resources. The higher the medical resources utilization rate, the lower the probability of an individual contracting a disease. The impact of medical resources on the infection rate of diseases (z) is shown in Fig. 7. Where $z = (1 - \varepsilon)^R$. The higher the utilization rate of medical resources ε , the greater the attenuation of the infection rate of medical resources. When the utilization rate of medical resources is fixed, the attenuation of the infection rate of medical resources will increase with the increase of the amount of medical resources owned by an individual.

Assumption 7. The impact of medical resources on disease recovery rate

The recovery rate only correlates with the quantity of medical resources possessed by individuals. Greater access to healing medical resources improves treatment efficacy, resulting in a higher recovery rate from diseases. So we assume that the probability of individual i transitioning from state A_1I to state S can be described as:

$$\mu_i^{A_1I}(t) = 1 - \mu_0 (1 - \varepsilon)^{R_i^{A_1I}(t)} \quad (25)$$

where $\mu_0 \in (0, 1]$ is a parameter characterizing the ease of recovery from disease. A larger μ_0 indicates a more challenging recovery process. As the value of μ_0 increases, the probability of recovery decreases. Additionally, ε represents the utilization rate of medical resources. As the utilization rate of medical resources increases, the recovery rate for individuals improves as shown in Fig. 8.

3. Theoretical analyses

In this section, we conduct a theoretical analysis of our model to derive the critical threshold of disease transmission by used the MMCA method. According to the description in the previous section, the N individuals can be in three states: US , A_1S , A_2S and A_1I . The probabilities that individual i being in these four states at time step t can be represented as $p_i^{US}(t)$, $p_i^{A_1S}(t)$, $p_i^{A_2S}(t)$, $p_i^{A_1I}(t)$, satisfying $p_i^{US}(t) + p_i^{A_1S}(t) + p_i^{A_2S}(t) + p_i^{A_1I}(t) = 1$.

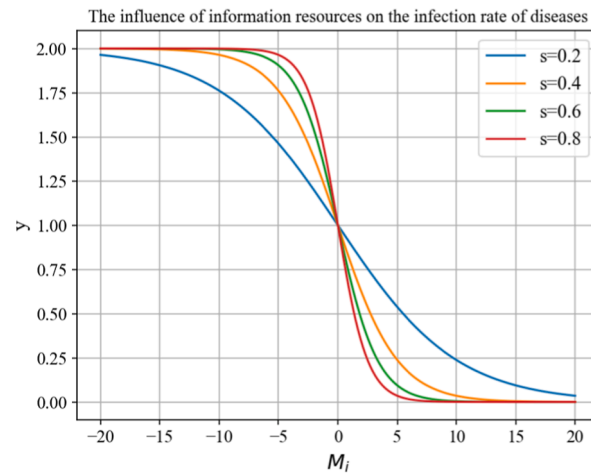


Fig. 6. The impact of information resources on the disease under different information resource sensitivities.

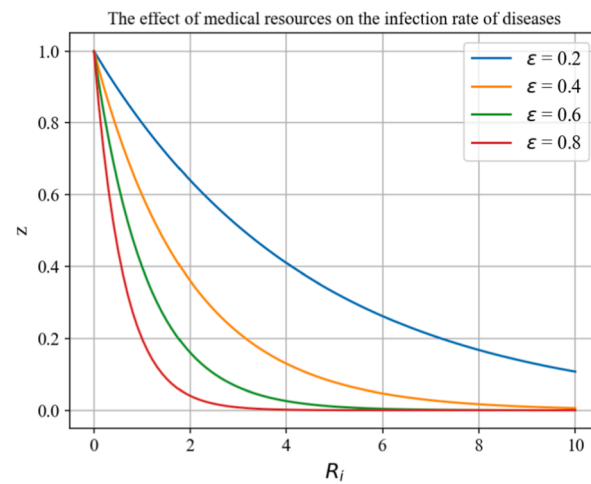


Fig. 7. The effect of medical resources on the infection rate of disease under different resource utilization.

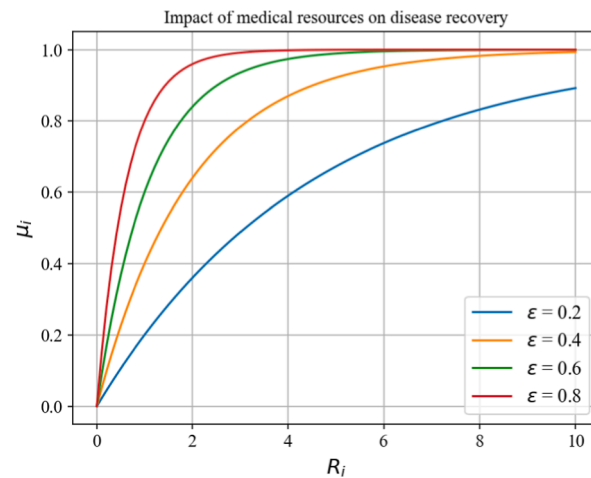


Fig. 8. When $\mu_0 = 1$, the impact of medical resources on disease recovery.

Let $A = (a_{ij})_{N \times N}$ and $B = (b_{ij})_{N \times N}$ be the adjacency matrix of awareness layer network and disease layer network, respectively. $r_i^{A_1}(t)$ and $r_i^{A_2}(t)$ indicate that node i in the unaware state at time step t has not shifted to A_1 or A_2 state in the awareness layer, respectively. Therefore, $r_i^{A_1}(t)$ and $r_i^{A_2}(t)$ can be characterized by the following formula:

$$r_i^{A_1}(t) = \prod_j [1 - a_{ji} P_j^{A_1}(t) \lambda_1] \quad (26)$$

$$r_i^{A_2}(t) = \prod_j [1 - a_{ji} P_j^{A_2}(t) \lambda_2] \quad (27)$$

Furthermore, the terms $\Theta_i^{A_1}(t)$ and $\Theta_i^{A_2}(t)$ are introduced to represent the probability that an unaware individual i transitions to A_1 state and A_2 state at time step t , respectively. The term $\Theta_i^U(t)$ is denoted as the probability that the unaware individual i remains in the unaware state at time step t . When individuals are exposed to both A_1 and A_2 individuals, they choose to trust A_1 first because of the psychological cost. If an individual chooses to believe A_2 not A_1 and then finds out that disease did is dangerous, they may feel regret or anxiety because they missed the opportunity to take protective measures accordingly. This psychological cost may prompt individuals to be more inclined to believe A_1 in order to avoid missing out on possible opportunities. Consequently, $\Theta_i^U(t)$, $\Theta_i^{A_1}(t)$, and $\Theta_i^{A_2}(t)$ can be characterized using the following formula:

$$\Theta_i^U(t) = r_i^{A_1}(t) r_i^{A_2}(t) \quad (28)$$

$$\Theta_i^{A_1}(t) = 1 - r_i^{A_1}(t) \quad (29)$$

$$\Theta_i^{A_2}(t) = 1 - r_i^{A_2}(t) - (1 - r_i^{A_1}(t)) (1 - r_i^{A_2}(t)) \quad (30)$$

We define $q_i^U(t)$, $q_i^{A_1}(t)$ and $q_i^{A_2}(t)$ respectively represent the probability that the US, A_1S, A_2S individual i is not infected at time step t . Therefore, $q_i^U(t)$, $q_i^{A_1}(t)$ and $q_i^{A_2}(t)$ can be characterized by the following formula:

$$q_i^U(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{US}(t)] \quad (31)$$

$$q_i^{A_1}(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{A_1S}(t)] \quad (32)$$

$$q_i^{A_2}(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{A_2S}(t)] \quad (33)$$

Then, in order to better understand the dynamic equation, we propose a discrete probability tree as shown in Fig. 9, which shows all possible transition processes between different states, illustrating the probability of transition between possible states after awareness diffusion and the transmission of the disease for each time step of individual i .

Based on the transition probability trees, we can develop following dynamic evolution equations:

$$p_i^{US}(t+1) = p_i^{US}(t) \Theta_i^U(t) q_i^U(t) + p_i^{A_1S}(t) \delta_1 q_i^U(t) + p_i^{A_2S}(t) \delta_2 q_i^U(t) + p_i^{A_1I}(t) \delta_1 \mu_i(t) \quad (34)$$

$$p_i^{A_1S}(t+1) = p_i^{US}(t) \Theta_i^{A_1}(t) q_i^{A_1}(t) + p_i^{A_1S}(t) (1 - \delta_1) (1 - \eta_1) q_i^{A_1}(t) + p_i^{A_2S}(t) (1 - \delta_2) \eta_2 q_i^{A_1}(t) + p_i^{A_1I}(t) (1 - \delta_1) (1 - \eta_1) \mu_i \quad (35)$$

$$p_i^{A_2S}(t+1) = p_i^{US}(t) \Theta_i^{A_2}(t) q_i^{A_2}(t) + p_i^{A_1S}(t) (1 - \delta_1) \eta_1 q_i^{A_2}(t) + p_i^{A_2S}(t) (1 - \delta_2) (1 - \eta_2) q_i^{A_2}(t) + p_i^{A_1I}(t) (1 - \delta_1) \eta_1 \mu_i \quad (36)$$

$$p_i^{A_1I}(t+1) = p_i^{US}(t) \left[\Theta_i^U(t) (1 - q_i^U(t)) + \Theta_i^{A_1}(t) (1 - q_i^{A_1}(t)) + \Theta_i^{A_2}(t) (1 - q_i^{A_2}(t)) \right] + p_i^{A_1S}(t) \left[(1 - \delta_1) (1 - \eta_1) (1 - q_i^{A_1}(t)) + \delta_1 (1 - q_i^U(t)) + (1 - \delta_1) \eta_1 (1 - q_i^{A_2}(t)) \right] + p_i^{A_2S}(t) \left[(1 - \delta_2) \eta_2 (1 - q_i^{A_1}(t)) + \delta_2 (1 - q_i^U(t)) + (1 - \delta_2) (1 - \eta_2) (1 - q_i^{A_2}(t)) \right] + p_i^{A_1I}(t) [(1 - \delta_1) \eta_1 (1 - \mu_i) + \delta_1 (1 - \mu_i) + (1 - \delta_1) (1 - \eta_1) (1 - \mu_i)] \quad (37)$$

when the system tends to steady state, each term in Eq. (34)–(37) satisfies:

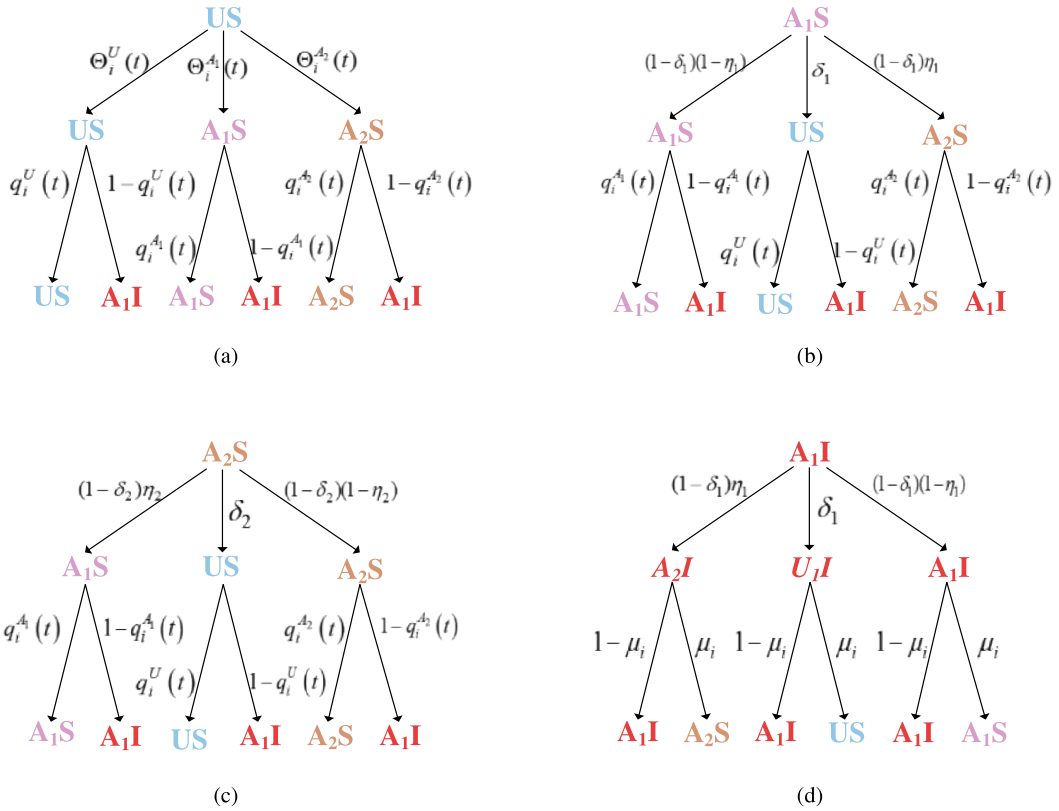


Fig. 9. The probability tree of four state changes.

$$p_i^{US}(t+1)_{t \rightarrow \infty} = p_i^{US}(t)_{t \rightarrow \infty} = p_i^{US} \quad (38)$$

$$p_i^{A_1S}(t+1)_{t \rightarrow \infty} = p_i^{A_1S}(t)_{t \rightarrow \infty} = p_i^{A_1S} \quad (39)$$

$$p_i^{A_2S}(t+1)_{t \rightarrow \infty} = p_i^{A_2S}(t)_{t \rightarrow \infty} = p_i^{A_2S} \quad (40)$$

$$p_i^{A_1I}(t+1)_{t \rightarrow \infty} = p_i^{A_1I}(t)_{t \rightarrow \infty} = p_i^{A_1I} \quad (41)$$

Next, we use the dynamic equation when $t \rightarrow \infty$ to solve the epidemic threshold. When β is infinitely close to β_c , the proportion of infected individuals in the system is infinitesimal, that is, $P_i^{A_1I} = \theta_i \ll 1$. Accordingly, ignoring the higher-order terms of $P_i^{A_1I}$, the approximate values of equations (31)–(33) are:

$$q_i^U(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{US}] \approx 1 - \beta_i^{US} \sum_j b_{ij} \theta_j \quad (42)$$

$$q_i^{A_1}(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{A_1S}] \approx 1 - \beta_i^{A_1S} \sum_j b_{ij} \theta_j \quad (43)$$

$$q_i^{A_2}(t) = \prod_j [1 - b_{ij} P_j^I(t) \beta_i^{A_2S}] \approx 1 - \beta_i^{A_2S} \sum_j b_{ij} \theta_j \quad (44)$$

For the convenience of derivation, we set $\sigma_i^{US} = \beta_i^{US} \sum_j b_{ij} \theta_j$, $\sigma_i^{A_1S} = \beta_i^{A_1S} \sum_j b_{ij} \theta_j$ and $\sigma_i^{A_2S} = \beta_i^{A_2S} \sum_j b_{ij} \theta_j$. Accordingly, by replacing these three expressions in formulas (42)–(44), we can get:

$$q_i^U(t) = 1 - \sigma_i^{US} \quad (45)$$

$$q_i^{A_1}(t) = 1 - \sigma_i^{A_1S} \quad (46)$$

$$q_i^{A_2}(t) = 1 - \sigma_i^{A_2S} \quad (47)$$

Then take formulas (45–47) into formulas (34–37), we obtain:

$$p_i^{US} = p_i^{US} \Theta_i^U (1 - \sigma_i^{US}) + p_i^{A_1S} \delta_1 (1 - \sigma_i^{US}) + p_i^{A_2S} \delta_2 (1 - \sigma_i^{US}) + p_i^{A_1I} \delta_1 \mu_i \quad (48)$$

$$p_i^{A_1S} = p_i^{US} \Theta_i^{A_1} (1 - \sigma_i^{A_1S}) + p_i^{A_1S} (1 - \delta_1) (1 - \eta_1) (1 - \sigma_i^{A_1S}) + p_i^{A_2S} (1 - \delta_2) \eta_2 (1 - \sigma_i^{A_1S}) + p_i^{A_1I} (1 - \delta_1) (1 - \eta_1) \mu_i \quad (49)$$

$$p_i^{A_2S} = p_i^{US} \Theta_i^{A_2} (1 - \sigma_i^{A_2S}) + p_i^{A_1S} (1 - \delta_1) \eta_1 (1 - \sigma_i^{A_2S}) + p_i^{A_2S} (1 - \delta_2) (1 - \eta_2) (1 - \sigma_i^{A_2S}) + p_i^{A_1I} (1 - \delta_1) \eta_1 \mu_i \quad (50)$$

$$p_i^{A_1I} = p_i^{US} [\Theta_i^U \sigma_i^{US} + \Theta_i^{A_1} \sigma_i^{A_1S} + \Theta_i^{A_2} \sigma_i^{A_2S}] + p_i^{A_1S} [(1 - \delta_1) (1 - \eta_1) \sigma_i^{A_1S} + \delta_1 \sigma_i^{US} + (1 - \delta_1) \eta_1 \sigma_i^{A_2S}] + p_i^{A_2S} [(1 - \delta_2) \eta_2 \sigma_i^{A_1S} + \delta_2 \sigma_i^{US} + (1 - \delta_2) (1 - \eta_2) \sigma_i^{A_2S}] + p_i^{A_1I} [(1 - \delta_1) \eta_1 (1 - \mu_i) + \delta_1 (1 - \mu_i) + (1 - \delta_1) (1 - \eta_1) (1 - \mu_i)] \quad (51)$$

By ignoring the higher-order terms in the first three expressions and ignoring the higher-order $O(\theta)$, we get:

$$p_i^{US} = p_i^{US} \Theta_i^U + p_i^{A_1S} \delta_1 + p_i^{A_2S} \delta_2 \quad (52)$$

$$p_i^{A_1S} = p_i^{US} \Theta_i^{A_1} + p_i^{A_1S} (1 - \delta_1) (1 - \eta_1) + p_i^{A_2S} (1 - \delta_2) \eta_2 \quad (53)$$

$$p_i^{A_2S} = p_i^{US} \Theta_i^{A_2} + p_i^{A_1S} (1 - \delta_1) \eta_1 + p_i^{A_2S} (1 - \delta_2) (1 - \eta_2) \quad (54)$$

We use the above three formulas instead of (51), we can get:

$$\mu_i \theta_j = \sigma_i^{US} p_i^{US} + \sigma_i^{A_1S} p_i^{A_1S} + \sigma_i^{A_2S} p_i^{A_2S} \quad (55)$$

We rewrite Eq. (55) as following form:

$$\sum_j \left\{ \left[\sigma_i^{US} p_i^{US} + \sigma_i^{A_1S} p_i^{A_1S} + \sigma_i^{A_2S} p_i^{A_2S} \right] b_{ij} - \frac{1 - \mu_0 (1 - \varepsilon) R_i^{A_1I}}{\beta_0} \zeta_{ij} \right\} \theta_j \approx 0 \quad (56)$$

where $\sigma_i^{US} = \frac{2}{1 + e^{-sM_i^{US}}} (1 - \varepsilon) R_i^{US}$, $\sigma_i^{A_1S} = \frac{2\gamma_1}{1 + e^{-sM_i^{A_1S}}} (1 - \varepsilon) R_i^{A_1S}$, $\sigma_i^{A_2S} = \frac{2\gamma_2}{1 + e^{-sM_i^{A_2S}}} (1 - \varepsilon) R_i^{A_2S}$, ζ_{ij} is the element of the identity matrix. Let each element of matrix H be $\left[\sigma_i^{US} p_i^{US} + \sigma_i^{A_1S} p_i^{A_1S} + \sigma_i^{A_2S} p_i^{A_2S} \right] b_{ij}$. Then, the solution to Eq. (56) can be transformed into solving the eigenvalue of matrix H . Characterizing $\Lambda_{\max}(H)$ as the maximum eigenvalue of matrix H , which is composed by $\left[\sigma_i^{US} p_i^{US} + \sigma_i^{A_1S} p_i^{A_1S} + \sigma_i^{A_2S} p_i^{A_2S} \right] b_{ij}$. It is closely linked to the network structure of the disease layer, information resource sensitivity and medical resource utilization. Finally, we can get the epidemic threshold of this model as:

$$\beta_c \approx \frac{1 - \mu_0 (1 - \varepsilon) R_i^{A_1I}}{\Lambda_{\max}(H)} \quad (57)$$

From Eq. (57), we can find that β_c is closely linked to the network structure of the second layer, information resource sensitivity s and medical resource utilization ε , the attenuation factor of infection rate γ_1 , enhancement factor γ_2 , amount of information resources M_i and medical resources R_i , and the dependence of disease recovery on medical resources μ_0 .

4. Numerical and simulations results

In this section, we conducted simulations of the awareness-resource-disease coevolution spreading model to investigate the impact of two competing information resources sharing and medical resources allocation on disease dynamics in multiplex networks. Thereinto, the networks utilized are Barabasi-Albert (BA) scale-free networks. And there are 1000 nodes in each network, with the mean degree of the awareness layer and disease layer meeting the specified criteria $< k_A >$

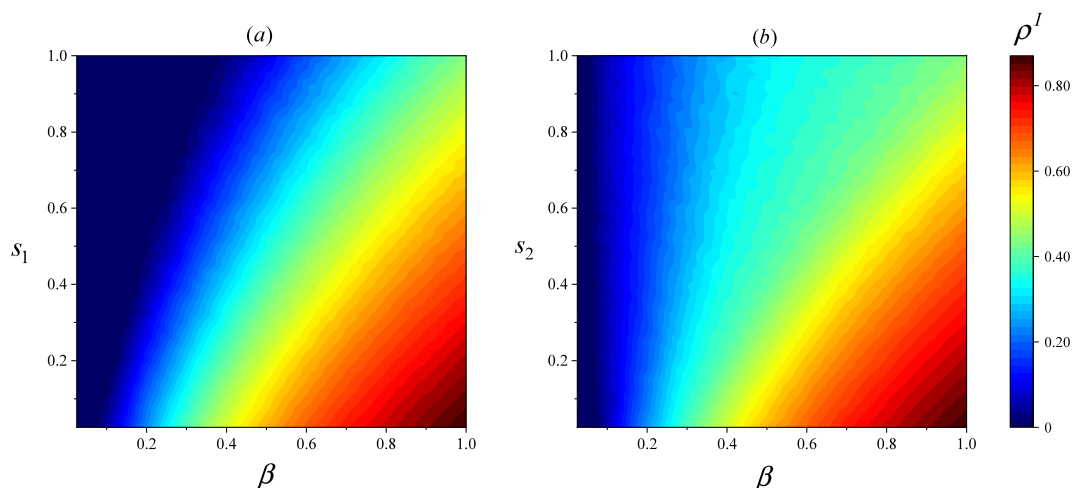


Fig. 10. ρ^I as a function of the infection rate influenced by varying sensitivities to different information resources. (a) Under the synergistic effect of information resources. (b) Under the hedging effect of information resources.

$= 6$, $\langle k_B \rangle = 3$ (Mossong et al., 2008). Additionally, the initial quantities of nodes A_1 and I were assumed to be 0.2 (Chinazzi et al., 2020), the initial quantities of nodes A_2 was set as 0.3.

Initially, we examined how the synergistic and hedging effects of information resources affects the scale of disease infection in Fig. 10. Panel (a) presents the synergy effect of positive information resources alone, showing that as the information sensitivity increases, the infection scale shows a monotonically decreasing trend - this indicates that the synergy effect of positive information resources can inhibit the spread of the disease. In contrast, panel (b) incorporates the hedging effect of negative information resources, revealing the effects of both the synergy and hedging of information resources on disease scale. Transitioning from (a) to (b), it is evident that the hedging effect of negative information resources against positive information resources enlarges the scale of infection, particularly during the early stages of an outbreak. However, as most of individuals become infected and increasingly perceive the impacts of the disease, the hedging effect of negative information resources is significantly diminished. In the later stages of disease progression, the hedging effect of information resources becomes less pronounced. These results highlight the dual necessity of controlling the epidemic: (i) In the crucial early stage (when the counter-effect reaches its peak), prompt intervention measures should be taken to counter false information resources; (ii) The sensitivity of information resources should be enhanced first to maximize the synergy effect of positive information resources in order to curb the spread of the epidemic.

Then, we detailed investigated the effect of the sensitivity of information resources on the scale of disease infection in Fig. 11. Panels (a) and (b) are heat maps and their corresponding line graphs. Both panels (a) and (b) illustrate that as the sensitivity of information resources increases, the final infection scale of the disease will decrease. Panel (a) Clearly demonstrates that when the sensitivity to information resources is extremely high, regardless of the infection rate, the overall infection scale will be controlled below half of the population. This indicates that when individuals closely monitor the

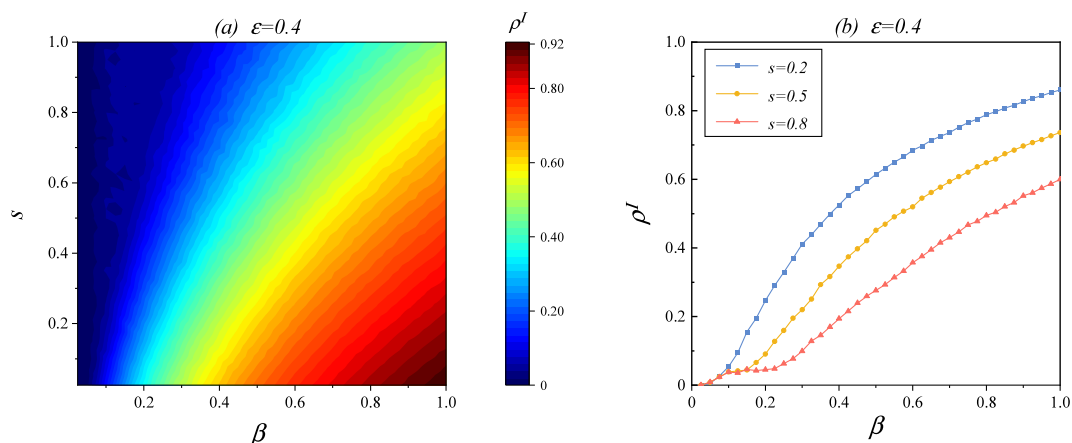


Fig. 11. ρ^I as a function of the infection rate influenced by varying sensitivities to different information resources.

information resources they receive, especially when there are more positive information resources than negative ones, they are more likely to take protective measures. Moreover, when individuals are highly sensitive to information resources, once the positive information resources are slightly more than the negative ones, there is a high probability that individuals will take protective measures, thereby significantly reducing the possibility of infection. (b) Reveals that increasing the sensitivity of information resources significantly reduces the degree of disease infection, but does not change the threshold for disease outbreak.

In Fig. 12, we analyzed the influence of medical resources on the scale of disease infection. In panel (a), it is evident that when the utilization rate of medical resources reaches a high level, the likelihood of a disease outbreak is significantly reduced or even eliminated. When medical resources are fully deployed—whether through comprehensive vaccination programs, timely treatments, or other preventive measures—the ability to effectively suppress or prevent disease outbreaks becomes significantly enhanced. For instance, children are routinely vaccinated at a young age to protect them from various diseases, and during an epidemic like COVID-19, individuals may receive multiple doses of the vaccine to prevent infection. This demonstrates that high medical resource availability fundamentally prevents outbreak escalation rather than merely mitigating outbreak scale. In (b), the utilization rate of medical resources is shown to play a crucial role in determining the threshold for disease outbreaks. Higher utilization rates are linked to increased outbreak thresholds. This indicates that when medical resources are more extensively used, there is a greater ability to prevent or control outbreaks before they become severe. For instance, in situations where medical resources are fully utilized (like widespread vaccination or timely medical care), diseases are less likely to become epidemics because of the effective containment and management strategies in place.

In order to further study the impact of the sensitivity of information resources and the utilization rate of medical resources on the scale of disease infection, we simulated the heat map of $s - \varepsilon$ as shown in Fig. 13. As depicted in (a) and (b), both increasing the utilization rate of medical resources and enhancing sensitivity to information resources result in a reduction in the final scale of infection. Specifically, while improving sensitivity to information resources shows a gradual

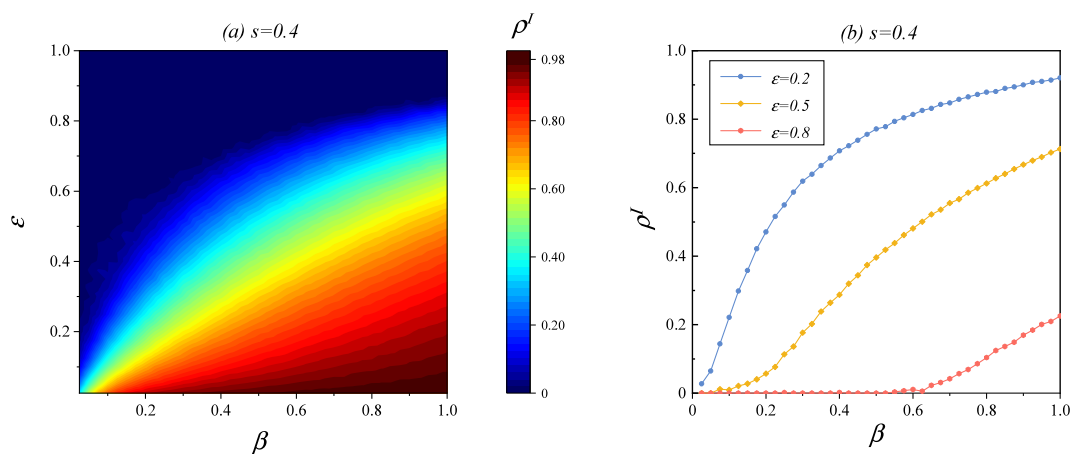


Fig. 12. ρ^I as a function of the infection rate β under the influenced by varying levels of medical resources utilization.

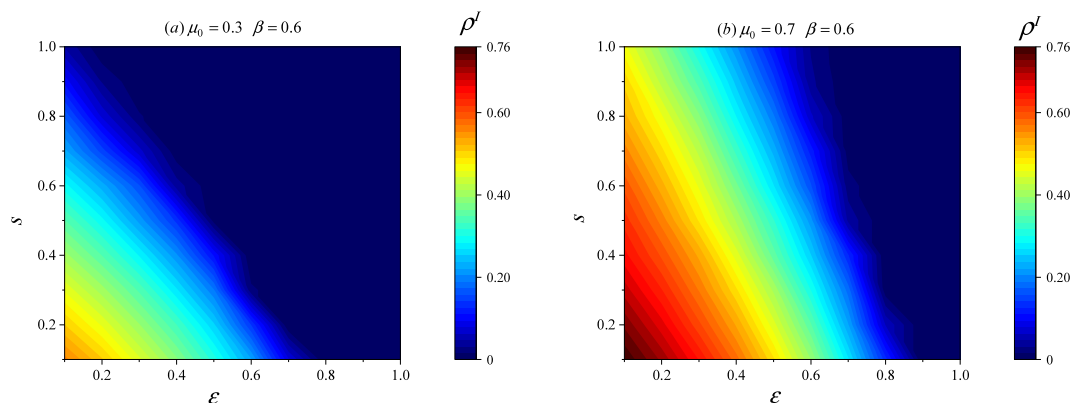


Fig. 13. ρ^I as a function of the infection rate β under the influence of different utilizations of medical resources and information resource sensitivities. (a) $\mu_0 = 0.3$ $\beta = 0.6$, (b) $\mu_0 = 0.7$ $\beta = 0.6$

decline in infection scale, a notable observation is that a low-to-high increase in medical resource utilization leads to a more rapid and substantial reduction to the lowest levels of infection. This highlights the profound discrepancy in the efficacy of medical resources compared to information resources. That is, compared with information resources, medical resources have a better inhibition effect in mitigating the spread of disease. Most importantly, integrating both information resources and medical resources yields optimal disease control. For example, when both information resources and medical resources are optimally coordinated ($s = 0.6$, $\mu = 0.6$) in panel (b), the heatmap shows infection density dropping to near-zero levels in the deep blue regions. By contrast, information resources alone ($s = 0.6$, $\mu = 0$) cannot suppress outbreaks (orange regions), nor can medical resources alone ($s = 0$, $\mu = 0.6$) achieve optimal control (green regions). Comparing (a) and (b) reveals a critical correlation: in scenarios where disease recovery heavily relies on medical resources ($\mu_0 = 0.7$), a lower utilization rate of these resources corresponds to a larger scale of infection. This underscores the importance of enhancing the utilization of medical resources in order to effectively combat the disease. When an individual's immune system is unable to adequately defend against the disease, it is crucial to make full use of available medical resources. In this context, measures such as increasing access to health services, promoting the regular use of masks and ensuring individual vaccination are key strategies to prevent infection. These measures to improve the utilization of health resources are essential to minimize disease transmission and strengthen public health resilience.

In Fig. 14, we conducted an analysis on how varying proportion of medical resources donation influence the scale of infection. It became evident that as the proportion of medical resources donation increases, there is a corresponding gradual decrease in the scale of disease infection. This suppression effect becomes especially pronounced at higher infection rates. This is because more individuals choose to donate their own medical resources to infected neighbors to accelerate recovery and thereby limit onward transmission. When the donation proportion is low, infection scales remain large and the transmission chain persists. Conversely, when the proportion increases to a moderate-to-high level, infection scales decline significantly: individuals donate medical resources (e.g., drugs and medical equipment) to nearby infected individuals. This reciprocal behavior promotes faster recovery while reducing the transmission risk posed by infected individuals, thereby effectively controlling epidemic spread. The simulation results remind us that in the face of highly transmissible infectious diseases, it is particularly important to establish an effective medical resource sharing mechanism and enhance the participation of community members, so as to help control infection and protect public health at critical moments.

Figs. 15 and 16 simulated the effect of the state of awareness on the scale of disease infection. We can see that as γ_1 decreases, the scale of infection decreases. This correlation arises because positive awareness improves understanding of how diseases are transmitted, the risk of infection, and the effectiveness of protective measures. When individuals have the right information, they are more likely to follow public health guidance and take effective measures such as vaccinations, social distancing, and frequent hand washing. This self-protective behavior can significantly reduce the rate of transmission within the community, ultimately reducing the overall number of infections. Conversely, as γ_2 increases the scale of infection increases. This is because misconceptions can lead individuals to ignore critical protective measures or adopt ineffective behaviors, thereby accelerating disease transmission. For example, some individuals may underestimate disease severity, neglect social distancing, or fail to wear masks—increasing infection risk for themselves and others. Particularly when infection rates surge, these misconceptions amplify epidemic spread and precipitate severe public health crises. Therefore, promoting accurate and comprehensive disease knowledge is crucial for effectively reducing transmission and mitigating population-level health impacts. In addition, the evolution of (a) through (c) shows that combining information resources with medical resources is significantly more effective in curbing the spread of disease, which is consistent with the conclusions stated earlier. Through accurate information resource sharing and necessary medical resource intervention,

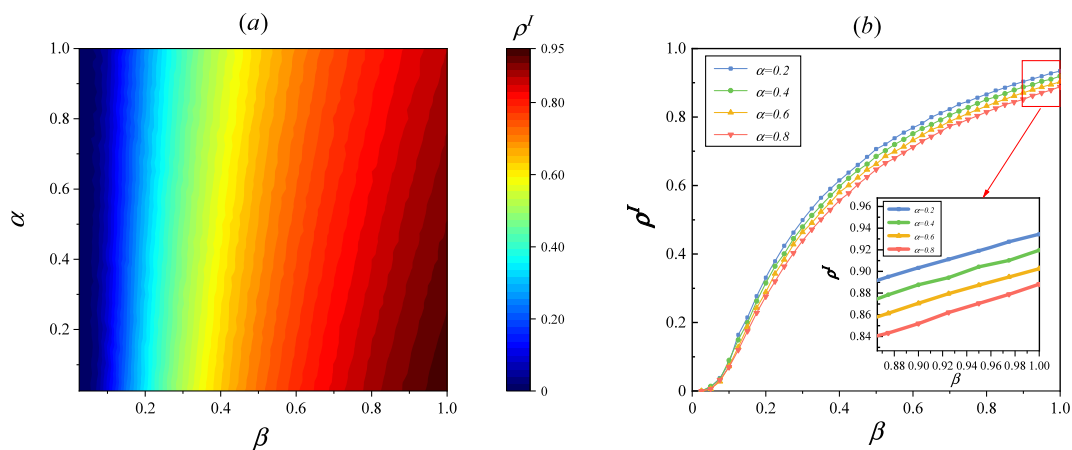


Fig. 14. ρ^I as a function of the infection rate β under the influence of different probability of medical resource donation. The remaining parameters are set to be $s = 0.2$, $\varepsilon = 0.4$

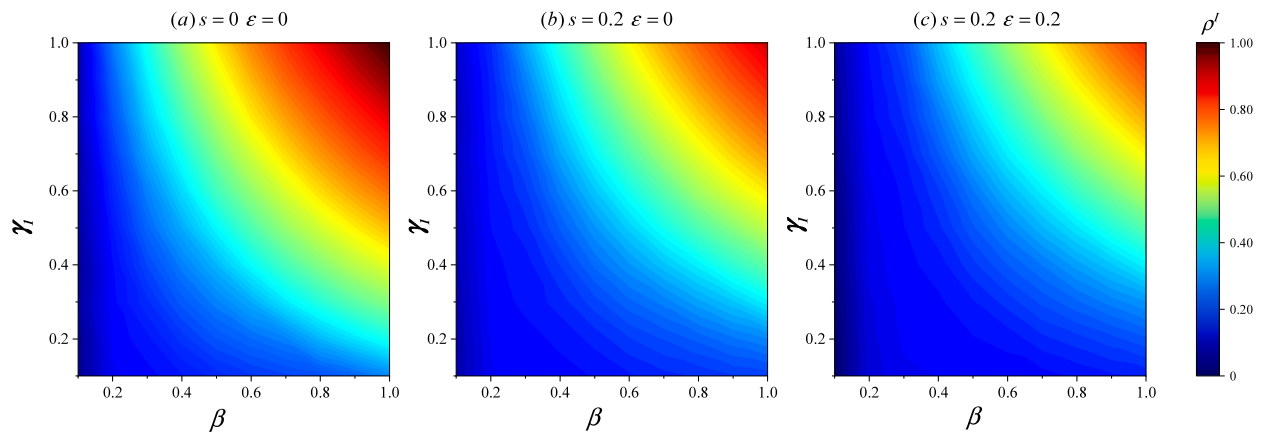


Fig. 15. Densities of infected individuals in the parameter space $\gamma_1 - \beta$. The remaining parameters are set to be $\lambda_1 = 0.1, \lambda_2 = 0.5, \alpha = 1, \eta_1^0 = \eta_2^0 = 0.1, \pi = 0.1, \varphi_1 = 0.5, \varphi_2 = 0.5, a = 1, \delta_1 = 0.5, \delta_2 = 0.1, \mu_0 = 0.5, \gamma_2 = 1.2$

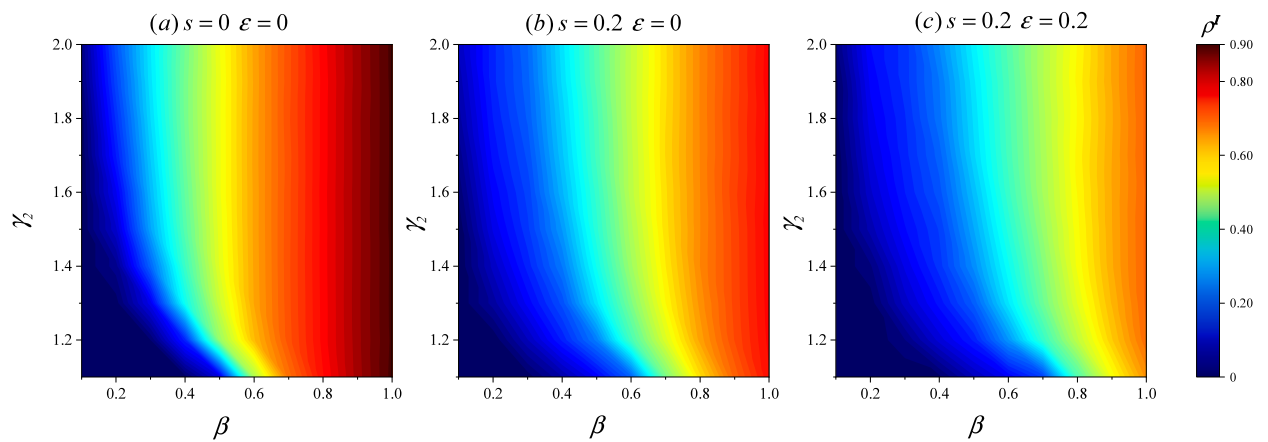


Fig. 16. Densities of infected individuals in the parameter space $\gamma_2 - \beta$. The remaining parameters are set to be $\lambda_1 = 0.1, \lambda_2 = 0.5, \alpha = 1, \eta_1^0 = \eta_2^0 = 0.1, \pi = 0.1, \varphi_1 = 0.5, \varphi_2 = 0.5, a = 1, \delta_1 = 0.5, \delta_2 = 0.1, \mu_0 = 0.5, \gamma_1 = 0.8$

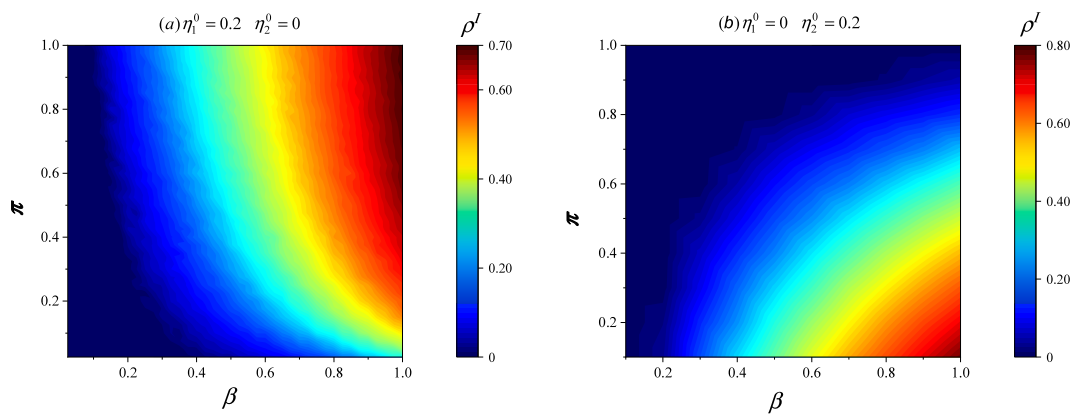


Fig. 17. Densities of infected individuals in the parameter space $\pi - \beta$. The remaining parameters are set to be $\lambda_1 = 0.1, \lambda_2 = 0.4, \alpha = 1, a = 1, \delta_1 = 0.4, \delta_2 = 0.2, \mu_0 = 0.5, \gamma_1 = 0.8, \gamma_2 = 1.2, s = 0.2, \epsilon = 0.2$.
(a) $\varphi_1 = 0, \varphi_2 = 1$ (b) $\varphi_1 = 1, \varphi_2 = 0$

this comprehensive strategy not only enhances public health preparedness, but also facilitates the collective response to the outbreak.

Fig. 17 analyzed the effect of the intensity of neighbor influence on the scale of disease infection. (a) Show the effect of neighbor influence intensity on the scale of disease infection when $A_1 \rightarrow A_2$. It can be seen that with the increase of π , the infection scale of the disease gradually increases. As the influence intensity of neighbors grows stronger, more A_1 individuals are converted to A_2 status, there is an increasing prevalence of misconceptions about diseases among the population, resulting in an increasing abundance of adverse information resources. Thus, individuals are encouraged to take inappropriate protective measures to increase the probability of infection and thus increase the scale of infection. (b) Show the effect of neighbor influence intensity on the scale of disease infection when $A_2 \rightarrow A_1$. It can be seen that with the increase of π , the infection scale of the disease gradually decreases. This is because with the increase of the intensity of the influence of A_1 neighbors, more and more individuals change from $A_2 \rightarrow A_1$ state, forming a correct cognition of the disease, and then generating positive information resources to share with others, prompting individuals to take the correct protective measures to protect themselves and thus reduce the infection rate of the disease.

5. Conclusion

In this paper, we proposed a coupled network model to characterize the complex interplay between awareness diffusion and disease transmission under the influence of synergistic-hedging effect of information resources and medical resources. Information is treated as a crucial yet intangible resource, with various types of information interacting to create either synergistic or hedging effects. When individuals receive the same type of information resources, a synergistic effect occurs due to mutual verification, which reinforces the behavior of adopting comprehensive protective measures against infection. Conversely, when individuals receive different types of information, a hedging effect arises due to mutual refutation, undermining the impact of the information. These synergistic and hedging effects determine the extent and efficacy of individuals' implementation of specific protective behaviors, ultimately influencing the dynamics of disease transmission. Additionally, we also considered that medical resources serve not only to treat diseases (such as medicines and vaccines) but also to prevent them during early outbreaks (including masks, protective clothing, and goggles), which also have an impact on disease. Based on this, we have developed a microscopic Markov chain theory to analyze the dynamics of the coevolution and have studied the influence of information resources and medical resources on disease transmission.

Monte Carlo simulations revealed that the hedging effect of negative information resources on positive information resources contributes to an increased scale of infection, particularly during the early stages of an outbreak. This finding suggests a critical policy imperative: public health authorities should prioritize rapid deployment of fact-based risk communication before misinformation gains traction. However, as the number of infected individuals rises, the hedging effect of negative information resources is substantially reduced. In the later stages of disease transmission, the impact of negative information resources becomes less significant, indicating that resource allocation should shift from intensive counter-misinformation campaigns toward scaling up medical interventions (e.g., testing, treatment) once community transmission is established. Besides, enhancing the sensitivity of information resources does not alter the disease outbreak threshold, but it significantly reduces the extent of disease infection. Specifically, although increasing sensitivity to information resources leads to a gradual decrease in infection scale, it is noteworthy that a shift from low to high medical resource utilization results in a faster and more significant reduction of infections to their minimal levels. So, making medical resource deployment the higher-leverage intervention when rapid containment is required. Furthermore, the integration of information resources with medical resources proves to be more effective in controlling disease. This indicates that public health strategies should carry out both information resources dissemination and medical resources allocation simultaneously, rather than focusing solely on one of them—the ideal control of an epidemic requires the implementation of both measures at the same time.

CRedit authorship contribution statement

Liang'an Huo: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Jinglin Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available within the article and its supplementary material.

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